International Journal of Information Security (2026) 25:172
https://doi.org/10.1007/s10207-026-01341-y

REGULAR CONTRIBUTION

The ICRIM framework: an integrated conceptual model of cyber resilience indicators

Vasiliki Tzavara[1] · **Savvas Vassiliadis**[1]

Received: 13 May 2026 / Accepted: 22 September 2026
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Abstract

Cyber resilience measurement remains conceptually fragmented across technical, organizational, and human-oriented research traditions. Existing approaches differ in terminology, level of analysis, form of operationalization, and evidence basis, while indicators are often not clearly distinguished from the metrics used to assess them. This study conducts a structured, theory-informed conceptual synthesis of 33 publications issued between 2011 and 2024 to examine how cyber resilience indicators have been conceptualized, operationalized, demonstrated, and validated. Using the a priori theory-informed structure described in the methodology, the analysis organizes the reviewed indicator families into three interdependent domains: technical, organizational, and socio-psychological and relational. Within the corpus, technical contributions typically use mathematical, temporal, structural, or probabilistic metrics. Organizational contributions rely mainly on maturity criteria, questionnaires, expert judgment, mission-oriented models, or hybrid approaches. Socio-psychological and relational constructs generally appear within broader cognitive, social, workforce, or organizational assessment structures rather than as standalone cyber resilience measures. Based on this synthesis, the study proposes the Integrated Cyber Resilience Indicators Model (ICRIM), a non-hierarchical conceptual structure for organizing heterogeneous indicator families, distinguishing indicators from operational metrics, and clarifying cross-domain relationships. ICRIM is not presented as a validated assessment instrument and does not prescribe universal metrics, weights, thresholds, or scoring procedures. Its contribution is to provide a transparent analytical basis for the future development and empirical testing of context-appropriate cyber resilience measures.

Keywords Cyber resilience · Resilience indicators · Cyber resilience measurement · Socio-technical systems · Organizational resilience · Socio-psychological and relational factors

1 Introduction

Cyber resilience has become an increasingly important concept in cybersecurity as digital systems, organizations, and essential services face persistent and evolving disruptions. Unlike conventional security approaches, which focus primarily on preventing or resisting attacks, cyber resilience concerns the ability to anticipate adverse events, withstand their effects, maintain or restore critical functions, and adapt following disruption [1, 2]. It therefore extends beyond technical protection and includes the organizational and human conditions through which systems are governed, operated, and restored.

Previous research has traced the conceptual development of cyber resilience and its expansion across systems engineering, organizational resilience, risk governance, and socio-technical research [3]. A related but distinct challenge concerns measurement. Although numerous frameworks, indicators, and metrics have been proposed, they differ substantially in their analytical level, terminology, methodological basis, and degree of operationalization.

Early work was predominantly engineering-oriented. The MITRE Cyber Resiliency Engineering Framework defined cyber resiliency through objectives such as anticipating, withstanding, recovering from, and evolving in response to adverse conditions, but did not initially specify a fixed metric set [1]. Subsequent MITRE publications introduced

✉ Vasiliki Tzavara
vtzavara@uniwa.gr

Savvas Vassiliadis
svas@uniwa.gr

[1] Department of Electrical and Electronics Engineering, University of West Attica, Athens, Greece

Published online: 07 October 2026

technical, cost-related, qualitative, quantitative, and semi-quantitative metrics intended to support systems-engineering and mission-level decisions [4–7].

Other approaches broadened the scope of assessment. The Resilience Matrix proposed by Linkov et al. [2] connected resilience stages with physical, information, cognitive, and social domains. Performance-based and control-theoretic studies introduced mathematical measures of degradation, retained functionality, recovery, controllability, observability, and systemic impact in cyber-physical environments [8–11]. More recent work has also incorporated mission requirements, governance arrangements, organizational maturity, workforce capabilities, and adaptive decision-making [12–14].

Despite these developments, three related problems remain. First, the terms indicator, metric, measure, objective, capability, and assessment criterion are often used inconsistently. Conceptual attributes of resilience are consequently not always distinguished from the operational procedures through which they are measured. Second, technical, organizational, and human-related approaches frequently employ different forms of evidence, including mathematical models, temporal measures, maturity assessments, expert judgment, questionnaires, and conceptual constructs. These approaches cannot be treated as methodologically equivalent. Third, constructs such as training, communication, trust, awareness, culture, and cooperation often cross analytical boundaries, creating uncertainty about whether they should be classified as organizational, behavioral, psychological, social, or relational.

This fragmentation limits conceptual clarity and complicates the development of coherent measurement approaches. The problem is not simply the absence of indicators. It is also the absence of a sufficiently explicit structure for distinguishing indicator families, identifying their form of operationalization, and examining their relationships across technical, organizational, and human or relational levels.

The present study addresses this problem through a structured, theory-informed conceptual synthesis of 33 publications issued between 2011 and 2024. It examines how cyber resilience has been represented through conceptual indicators, operational metrics, assessment frameworks, mathematical models, maturity structures, and applied measurement approaches.

The analysis addresses three questions:

1. What indicator families can be identified within the a priori technical, organizational, and socio-psychological and relational domains across existing cyber resilience approaches?
2. How have these indicator families been operationalized, demonstrated, or validated?
3. How can heterogeneous indicator traditions be organized within a coherent conceptual structure without treating them as methodologically equivalent?

To address these questions, the study proposes the Integrated Cyber Resilience Indicators Model (ICRIM). ICRIM is positioned at the indicator level rather than as a validated scoring or measurement instrument. It organizes cyber resilience indicators into three analytically distinct but interdependent domains: technical, organizational, and socio-psychological and relational. It also distinguishes conceptual indicators from operational metrics and identifies constructs whose meaning or assessment spans more than one domain.

The contribution of the study is threefold. First, it provides a transparent analytical classification of the conceptual and measurement approaches represented in the reviewed corpus. Second, it clarifies the distinction between indicators and metrics and compares differences in operationalization, evidence basis, application level, and validation status. Third, it proposes ICRIM as a non-hierarchical conceptual integration structure for relating technical performance, organizational capability, and socio-psychological and relational conditions.

ICRIM does not replace established engineering, maturity-assessment, or organizational resilience frameworks. Nor does the present study claim empirical validation, universal comparability, regulatory alignment, or a standardized scoring procedure. Its purpose is to establish a traceable conceptual basis for the subsequent development and testing of context-appropriate cyber resilience measures.

2 Methodology

2.1 Research design and scope

This study adopts a structured, theory-informed conceptual synthesis to examine and integrate existing approaches to the measurement of cyber resilience. Rather than conducting an exhaustive systematic review or generating primary empirical data, the study seeks to clarify the conceptual attributes that have been treated as indicators of cyber resilience and to examine how these attributes have been operationalized across different research traditions. This design is appropriate for a field characterized by conceptual heterogeneity, uneven levels of operationalization, and the coexistence of engineering, organizational, and human-oriented approaches to measurement [15, 16].

The temporal scope covers publications issued between 2011 and 2024. The lower boundary corresponds to the publication of the MITRE Cyber Resiliency Engineering Framework in 2011, which was the earliest source identified

in the analytical corpus to formulate an explicit engineering framework for cyber resiliency. The upper boundary was set at the end of 2024 because source identification, full-text review, data extraction, and analytical synthesis were conducted during 2025. The study therefore does not claim coverage of literature published after 2024.

The analysis was guided by an initial socio-technical structure comprising technical, organizational, and socio-psychological and relational domains. These domains were specified a priori as a theory-informed analytical lens rather than generated through a purely inductive coding process. They were not treated as fixed or mutually exclusive categories. Instead, the reviewed sources were compared iteratively to identify and refine the indicator families within this structure and to clarify their content, boundaries, overlaps, and cross-domain interdependencies.

A central methodological distinction is made between indicators and metrics. Indicators refer to the conceptual attributes, capacities, or conditions that signal the resilience of a system or organization, whereas metrics refer to the operational procedures, scales, formulas, or quantified expressions used to assess those indicators [17, 18]. Some reviewed sources therefore contribute explicit qualitative, quantitative, semi-quantitative, or hybrid metrics, while others contribute conceptual frameworks, objectives, principles, or assessment structures from which relevant indicators can be identified.

ICRIM is positioned primarily at the indicator level. It does not introduce a validated scoring procedure, weighting system, or new set of operational metrics. Instead, it proposes a conceptual structure for organizing heterogeneous indicator families, clarifying their relationships, and supporting their future operationalization through context-appropriate measurement methods. The study is therefore integrative rather than evaluative: it does not test the predictive validity, comparative performance, or practical effectiveness of the proposed model.

2.2 Search strategy and corpus construction

Source identification was conducted during the first quarter of 2025 and was completed by the end of March 2025 through searches in Scopus and Google Scholar, supplemented by direct searches of relevant institutional websites, including those of MITRE and the World Economic Forum. The principal search terms were "cyber resilience metrics," "cyber resiliency indicators," "cyber resilience measurement," and "cyber resilience framework." Backward reference searching was also undertaken by examining the reference lists of relevant publications to identify additional academic and institutional sources.

Results obtained through the different search routes were consolidated. Where the same publication was identified through more than one route, it was retained only once in the candidate source collection. Initial relevance screening was based on titles and abstracts. Records were excluded when cyber resilience was mentioned only incidentally or when the publication addressed a different research problem without substantive relevance to cyber resilience indicators, operational metrics, measurement approaches, or assessment frameworks. Where relevance could not be determined from the title and abstract alone, the full text was examined before an inclusion decision was made.

Source selection was relevance-based and guided by three analytical criteria. First, a publication had to address explicitly, or contribute substantively to, cyber resilience indicators, operational metrics, measurement approaches, or assessment frameworks. Second, it had to provide a theoretical, methodological, or modeling basis related to resilience engineering, socio-technical systems, adaptive capacity, organizational resilience, or the performance and recovery of cyber or cyber-physical systems. Third, a publication had to contribute to at least one of the levels of analysis relevant to the study: technical system performance, organizational governance and adaptive capability, or socio-psychological, behavioral, or relational conditions. Individual publications were not required to address all three levels.

The search was designed to support conceptual integration rather than exhaustive bibliometric coverage. Selection was based on the stated criteria rather than on a preregistered review protocol or a predefined exclusion checklist. Full texts were available for all included publications, and no source was retained solely on the basis of an abstract or a secondary description. Information concerning the existence and number of metrics, their form of operationalization, methodological basis, validation or demonstration procedures, simulations, and case applications was checked against the full text of each source.

Following the search and relevance-screening process, 33 publications met the study's selection criteria and were retained for analytical processing. No predetermined target number of publications or formal saturation threshold was used to determine the size of the corpus. The resulting corpus provided the conceptual breadth required for the intended synthesis because it included publications addressing the three a priori analytical domains and represented different forms of conceptualization, operationalization, and assessment relevant to the research questions. This assessment concerns the suitability of the resulting corpus for the intended conceptual synthesis and does not imply exhaustive coverage of the wider cyber resilience literature.

The corpus comprised peer-reviewed journal articles, conference papers, technical reports, institutional frameworks, and theoretical or modeling contributions. These publications constitute the complete corpus used for the comparative analysis and conceptual classification.

2.3 Data extraction and analytical classification

A structured analytical record was developed for each of the 33 publications. The extraction process documented: (a) whether the source introduced explicit cyber resilience metrics, proposed conceptual indicators, or provided a broader assessment framework; (b) the number or scope of the metrics, where specified; (c) the qualitative, quantitative, semi-quantitative, or hybrid form of operationalization; (d) the theoretical or methodological foundation of the approach; (e) whether the contribution remained conceptual or was examined through mathematical modeling, simulation, case-based demonstration, or practical application; (f) its relationship to an existing resilience, cybersecurity, or systems-engineering framework; and (g) its principal contribution to the conceptualization or measurement of cyber resilience. The extracted information was checked against the full text of the respective publication.

The terminology used by the original sources was retained during the initial extraction. Concepts described as goals, objectives, principles, capabilities, indicators, measures, or metrics were therefore not automatically treated as equivalent. For the purposes of the synthesis, a construct was treated as an indicator when it represented an attribute, condition, capacity, or dimension relevant to cyber resilience. It was treated as a metric when the source specified an operational procedure, scale, formula, scoring method, or other rule through which that attribute could be assessed. Sources that did not introduce operational metrics could therefore still contribute to the identification and organization of relevant indicator families.

The analytical classification followed a theory-informed, deductive–iterative process. Technical, organizational, and socio-psychological and relational domains constituted the initial analytical structure through which the extracted material was examined. The three domains served as an organizing lens rather than as categories claimed to have emerged entirely inductively from the corpus.

The classification was refined through repeated comparison of the extracted constructs across the selected publications. Conceptually related constructs were grouped into broader indicator families, and their content, boundaries, and interdependencies were examined across sources. Technical indicators were associated primarily with system behavior, architecture, detection, degradation, continuity, redundancy, recovery, and operational performance. Organizational indicators concerned governance, institutional responsibilities, resource allocation, incident-management processes, communication structures, preparedness, training provision, and organizational learning. Socio-psychological and relational indicators included cognitive and emotional responses, risk perception, awareness, behavioral adaptation, trust, cooperation, and relational or collective responses to cyber disruption.

The domains were not treated as mutually exclusive. Where a construct related to more than one domain, it was classified according to its principal object and mechanism of assessment, while its cross-domain relevance was retained. For example, the institutional provision of training was treated primarily as an organizational indicator, whereas changes in awareness, confidence, or protective behavior resulting from training were treated as socio-psychological or behavioral indicators. Organizational culture and formal communication arrangements were located primarily within the organizational domain, while interpersonal trust, shared sense-making, and cooperative behavior were considered relational components of the socio-psychological and relational domain. Institutional trust was treated as a cross-domain construct when it reflected both perceptions of legitimacy and the organizational practices shaping those perceptions.

Following classification, the sources were compared in relation to their indicator domain, form of operationalization, methodological basis, level of application, reported validation or demonstration status, and principal limitation. This comparative mapping was used to identify areas of convergence, conceptual overlap, and uneven operational development across the literature. The purpose was not to calculate the frequency or effectiveness of individual metrics, but to examine how different conceptual and measurement traditions could be organized within a coherent multi-domain conceptual structure. Table 1 provides a condensed overview of selected sources illustrating the principal traditions represented in the corpus. The complete analytical mapping of all 33 publications is provided in Supplementary Table S1.

2.4 Comparative frameworks and methodological boundaries

The comparative positioning of ICRIM initially drew on frameworks and measurement approaches included in the 33-source corpus, particularly the MITRE cyber resiliency framework and metrics publications, the Resilience Matrix proposed by Linkov et al. [2], and the World Economic Forum's Cyber Resilience Index. These sources represent different traditions, ranging from systems-engineering objectives and operational metrics to multidimensional matrices and organizational assessment structures.

A supplementary comparative examination was undertaken to clarify the relationship between ICRIM and three established institutional approaches that were not included as separate sources in the analytical corpus: NIST SP 800-160 Volume 2 Revision 1, the ENISA CSIRT Maturity Framework, and the CISA Cyber Resilience Review [25–27].

Table 1 Condensed comparative overview of selected cyber resilience conceptual and measurement approaches

Source	Contribution type	Primary indicator domain(s)	Form of operationalization	Evidence or application basis	Principal contribution and limitation
Bodeau and Graubart [1], Cyber Resiliency Engineering Framework	Conceptual engineering framework	Primarily technical; system and mission levels	Conceptual; no explicit metrics	Theoretical framework	Defines the foundational goals of anticipating, withstanding, recovering, and evolving. It provides an engineering structure but does not operationalize it through specific metrics
MITRE [4], Cyber Resiliency Metrics, Version 1.0, Rev. 1	Operational metrics framework	Primarily technical; organizational decision support	Qualitative, quantitative, and semi-quantitative	Metric templates and illustrative applications	Introduces 19 technical and 11 cost metrics linked to cyber resiliency objectives. The metrics require tailoring to specific systems, missions, and decision contexts
Ford et al. [19], Toward Metrics for Cyber Resilience	Conceptual and methodological critique	Cross-domain	Conceptual; emphasis on ordinal assessment	Theoretical analysis	Questions the feasibility of universal resilience metrics and emphasizes the contextual nature of measurement. It does not propose a specific operational metric set
Linkov et al. [2], Resilience Metrics for Cyber Systems	Multidimensional indicator and metric framework	Cross-domain: physical, information, cognitive, and social	Qualitative and quantitative	Conceptual matrix informed by literature and expert judgment	Organizes resilience across four stages and four domains. The proposed measures are illustrative and are not validated as a unified assessment system
Khan et al. [20], Cyber Resilience-by-Construction: Modeling, Measuring and Verifying	Formal resilience modeling framework	Technical	Quantitative and qualitative modeling	Formal modeling and firewall case demonstration	Links resilience to proactive, resistive, and reactive system behavior. Its application remains primarily technically oriented and system-specific

Table 1 (continued)

Source	Contribution type	Primary indicator domain(s)	Form of operationalization	Evidence or application basis	Principal contribution and limitation
Friedberg et al. [8], Towards a Resilience Metric Framework for Cyber-Physical Systems	Performance-based metric framework	Technical; cyber-physical systems	Quantitative	Mathematical formulation	Measures resilience through the deviation between nominal and actual system performance. It provides precise quantification but does not incorporate organizational or human conditions
MITRE [5], Cyber Resilience Metrics: Key Observations	Methodological guidance for metric development	System, mission, organizational, and broader levels	Qualitative, quantitative, and semi-quantitative	Conceptual and practice-oriented guidance	Clarifies principles and practical challenges in defining and applying resilience metrics. It does not provide a fixed or exhaustive metric set
Haque et al. [21], Cyber Resilience Framework for Industrial Control Systems: Concepts, Metrics, and Insights	Hierarchical resilience assessment framework	Technical, physical, and organizational	Hybrid; expert judgment and AHP-based quantification	Conceptual development and case demonstration	Adapts robustness, redundancy, resourcefulness, and rapidity to industrial control systems. Results depend on expert assessments and weighting decisions
MITRE [7], Cyber Resiliency Metrics and Scoring in Practice: Use Case Methodology and Examples	Scoring methodology and applied guidance	Primarily technical; system and mission levels	Qualitative, quantitative, and semi-quantitative	Use cases and applied scoring examples	Demonstrates how resilience measures can be selected and scored in specific contexts. The methodology requires extensive contextual tailoring
MITRE [6], Cyber Resiliency Metrics Catalog	Extensive operational metrics catalog	Primarily technical; system and mission levels	Qualitative, quantitative, and semi-quantitative	Structured metric catalog	Links nearly 500 representative metrics to resilience objectives, techniques, and design principles. It provides extensive operational detail but limited treatment of socio-psychological constructs
Hossain-McKenzie et al. [9], Performance-Based Cyber Resilience Metrics: An Applied Demonstration Toward Moving Target Defense	Performance-based metric set	Technical; cyber-physical and energy systems	Quantitative	Mathematical formulation and applied demonstration	Introduces Systemic Impact, Total Recovery Effort, and a Resilience Index. The measures remain dependent on system-specific performance variables

Table 1 (continued)

Source	Contribution type	Primary indicator domain(s)	Form of operationalization	Evidence or application basis	Principal contribution and limitation
Barbeau et al., Metrics to Enhance the Resilience of Cyber-Physical Systems [10]	Control-theoretic resilience metrics	Technical; cyber-physical systems	Quantitative	Mathematical modeling and simulation	Introduces steerability and monitorability metrics to assess structural resilience against covert attacks. The approach is technically specific and validated mainly through modeling
Segovia et al. [11], Cyber-Resilience Evaluation of Cyber-Physical Systems	Performance and structural metric set	Technical; cyber-physical systems	Quantitative	Mathematical modeling and numerical simulation	Proposes ten metrics covering performance degradation, recovery, controllability, and observability. Organizational and human dimensions remain outside its scope
Ahn et al. [22], Research on Improving Cyber Resilience by Integrating the Zero Trust Security Model with the MITRE ATT&CK Matrix	Operational performance metrics	Technical and operational	Quantitative	Experimental or scenario-based application	Uses time- and efficiency-based measures, including MTTA, MTTD, MTTR, detection rate, and recovery effectiveness. The focus remains primarily on technical response performance
Alevizos and Ta [23], TTP-Based Cyber Resilience Index	Probabilistic composite resilience index	Technical and organizational defense effectiveness	Quantitative	Probabilistic modeling using threat intelligence and POMDPs	Integrates threat behavior and defense effectiveness into a unified index. Its results depend on model assumptions, input quality, and contextual calibration
Alhidaifi et al. [24], A Survey on Cyber Resilience: Key Strategies, Research Challenges, and Future Directions	Review and synthesis	Cross-domain, with emphasis on technical recovery and adaptation	General quantitative and framework-based approaches	Literature survey	Consolidates existing measurement approaches and identifies standardization and implementation challenges. It does not introduce or validate a unified operational framework

Table 1 presents selected sources illustrating the principal conceptual and measurement traditions represented in the analytical corpus. It does not reproduce the complete corpus. The full source-level classification of all 33 publications is provided in Supplementary Table S1

These frameworks were selected because they represent complementary systems-engineering, capability-maturity, and operational-resilience perspectives.

The comparison considered the principal object and level of assessment adopted by each framework, the distinction between conceptual objectives or capabilities and their operational assessment, the treatment of organizational and human factors, and the role assigned to feedback, learning, and adaptation. Its purpose was not to evaluate the relative effectiveness of the frameworks or to suggest that ICRIM replaces their established procedures. It was used to clarify the relationship between ICRIM and existing approaches and to define more precisely the conceptual function that the proposed model is intended to perform.

These three frameworks were treated as supplementary comparative sources. They were not included in the 33-publication analytical corpus and were not used as sources from which the initial three-domain structure was derived. Their role was limited to positioning ICRIM in relation to established systems-engineering, maturity-assessment, and operational-resilience approaches.

Several methodological boundaries should be considered when interpreting the findings. The study employed a relevance-based source-selection process within a theory-informed conceptual synthesis and was not designed as an exhaustive systematic review. Although searches were conducted across academic search platforms, institutional websites, and the reference lists of relevant publications, no preregistered protocol, formal bibliometric procedure, or predetermined exclusion checklist was applied. The corpus is therefore intended to represent the principal conceptual and measurement approaches relevant to the study's purpose, but it cannot be considered a complete enumeration of the literature published during the period examined.

The source review, data extraction, and primary analytical classification were conducted by the first author. The resulting classification represents an interpretive synthesis and was not subjected to independent double coding or statistical assessment of inter-coder agreement. In addition, the three domains were used as an initial theory-informed structure rather than generated through purely inductive coding. Alternative classifications of individual cross-domain constructs may therefore remain possible.

Finally, the study does not empirically validate ICRIM or establish operational metrics, weighting procedures, scoring thresholds, or evidence of cross-organizational comparability. The supplementary comparison with NIST, ENISA, and CISA does not imply formal alignment, equivalence, or regulatory compliance. ICRIM is proposed as a conceptual structure for organizing heterogeneous indicator families and making their relationships explicit. Its measurement validity, practical applicability, and compatibility with particular sectoral or institutional assessment frameworks remain subjects for future empirical research.

3 Findings: indicator domains and patterns of operationalization

Using the initial theory-informed analytical structure described in Sect. 2.3, the comparative analysis organized the indicators reported in the 33 publications within three interrelated domains: technical, organizational, and socio-psychological and relational. The analysis refined their substantive content, clarified areas of overlap, and identified differences in how indicators were operationalized across the reviewed sources.

The corpus shows uneven levels of operational development across the three domains. Within the corpus, technical resilience is predominantly represented through explicit performance measures, mathematical formulations, and time-based or structural metrics. Organizational resilience is more commonly assessed through qualitative, semi-quantitative, or hybrid approaches involving governance arrangements, maturity criteria, expert judgment, and mission-oriented assessment. Socio-psychological and relational conditions are less often operationalized as standalone cyber resilience metrics and are generally incorporated into broader cognitive, social, workforce, cultural, or organizational assessment structures.

These patterns refer specifically to the resulting analytical corpus. They do not establish the frequency or effectiveness of indicator types across the entire literature. Instead, they show how different conceptual and measurement traditions contribute to the proposed taxonomy and where further operational development remains necessary.

3.1 Technical indicators of cyber resilience

Technical indicators constitute the most operationally developed domain within the reviewed corpus. They assess how systems, infrastructures, and cyber-physical processes perform before, during, and after adverse cyber events. Recurring indicator families include performance degradation, retained functionality, availability, detection and response time, recovery duration, redundancy, diversity, controllability, observability, systemic impact, recovery effort, and defense effectiveness [4, 6, 7, 9–11, 22].

The operationalization of these indicators ranges from descriptive or semi-quantitative scales to mathematical functions, temporal measures, probabilistic indices, and composite scores. Several studies examine their application through simulations, formal models, experimental scenarios, or case-based demonstrations. This gives technical indicators

relatively explicit measurement procedures and facilitates their application within defined system environments.

However, their interpretation remains dependent on system architecture, mission requirements, performance thresholds, threat scenarios, and measurement boundaries. Metrics developed for a power system, industrial control environment, autonomous system, or Zero Trust architecture cannot automatically be compared without common definitions and assumptions. Technical indicators also provide limited information about the governance, institutional, and human processes through which technical resilience is implemented and sustained.

3.2 Organizational indicators of cyber resilience

Organizational indicators concern the structures and capabilities through which resilience is governed, coordinated, and maintained. The reviewed sources associate this domain with governance responsibilities, resource allocation, mission prioritization, preparedness, incident-management processes, business continuity, communication arrangements, training provision, workforce capability, and organizational learning [5, 12–14, 28].

These indicators are commonly operationalized through maturity criteria, structured questionnaires, expert assessments, weighted scores, mission-oriented models, or combinations of qualitative and quantitative evidence. Unlike technical metrics, which may be derived directly from system performance data, organizational indicators frequently require an assessment of whether practices are formally defined, implemented, coordinated, reviewed, and improved.

Their dependence on judgment does not make them inherently less meaningful, but it requires transparent assessment criteria, evidence requirements, and scoring rules. Organizational size, sector, regulatory environment, leadership structure, and institutional maturity may also influence their interpretation. This domain therefore provides the institutional context within which technical information is converted into decisions, resources, coordinated action, and post-incident learning.

3.3 Socio-psychological and relational indicators of cyber resilience

The socio-psychological and relational domain includes individual, behavioral, interpersonal, and collective conditions that influence responses to cyber disruption. Relevant indicator families include awareness, risk perception, confidence, protective behavior, compliance, interpersonal trust, cooperation, shared sense-making, and behavioral adaptation.

This domain does not treat every social or institutional phenomenon as an individual psychological variable. Formal communication arrangements, professional culture, leadership practices, and organizational legitimacy are classified primarily as organizational constructs. Interpersonal trust, cooperation, shared interpretation, and behavioral responses are treated as relational or socio-psychological constructs. Institutional trust may cross both domains because it reflects perceptions of legitimacy as well as the organizational practices through which those perceptions are formed.

Within the reviewed corpus, these constructs are generally represented through broader cognitive and social domains, workforce and leadership capabilities, organizational culture, information flows, training, and awareness practices rather than through independently validated cyber resilience metrics [2, 12, 14, 29]. The corpus therefore supports their conceptual relevance but provides limited evidence of standardized operationalization, psychometric validation, or cross-context comparability. In ICRIM, they are consequently treated as indicator families requiring further empirical specification rather than as established measurement scales.

3.4 Cross-domain interdependencies and analytical synthesis

The classification also shows that several resilience constructs cannot be assigned exclusively to a single domain. Technical performance becomes organizationally meaningful when it is interpreted against mission priorities, continuity requirements, and decision thresholds. Organizational arrangements influence how technical information is communicated, how resources are mobilized, and how recovery actions are coordinated. Human and relational conditions affect whether procedures are understood, trusted, and implemented effectively.

For example, the institutional provision of training is an organizational indicator, whereas changes in awareness or protective behavior resulting from training belong primarily to the socio-psychological and relational domain. Formal communication structures are organizational, while shared sense-making and cooperation during an incident are relational. Similarly, retained system functionality is a technical outcome, but its significance depends on organizationally defined mission requirements.

The resulting taxonomy therefore does not represent three isolated or additive lists of indicators. It provides a structure for examining the cross-domain connections among technical performance, organizational capability, and socio-psychological and relational conditions. The synthesis identified recurring patterns of interdependence across these areas. Building on these patterns, ICRIM proposes specific relationships among the three domains as theory-informed conceptual propositions introduced by the authors. These

Table 2 Summary of indicator families and patterns of operationalization within the three a priori analytical domains

Indicator domain	Indicative indicator families	Typical operationalization in the corpus	Evidence or application pattern	Main analytical boundary	Representative sources
Technical	Performance degradation; retained functionality; availability; detection, response, and recovery time; redundancy and diversity; controllability and observability; systemic impact; recovery effort; defense effectiveness	Quantitative, mathematical, temporal, probabilistic, structural, and composite metrics; some qualitative or semi-quantitative scales	Formal modeling, simulation, experimental scenarios, metric catalogs, and case-based demonstrations	Strong dependence on system architecture, threat scenario, performance thresholds, and sectoral context; limited representation of organizational and human processes	MITRE [4, 6, 7]; Hossain-McKenzie et al. [9]; Barbeau et al. [10]; Segovia et al. [11]; Ahn et al. [22]; Alevizos and Ta [23]
Organizational	Governance and role clarity; resource allocation; mission prioritization; preparedness; incident management; communication structures; business continuity; training provision; workforce capability; organizational learning	Qualitative, semi-quantitative, and hybrid assessment; maturity criteria, questionnaires, expert judgment, weighted scoring, and mission-oriented modeling	Institutional frameworks, practice-oriented guidance, assessment tools, use cases, and simulations	Context dependence, judgment-based assessment, and limited standardization of evidence, weighting, and scoring; overlap with behavioral and relational conditions	MITRE [5]; Haque et al. [28]; Snyder et al. [12]; Llansó and McNeil [13]; World Economic Forum [14]; Al-Hawamleh [29]
Socio-psychological and relational	Awareness; risk perception; confidence; protective behavior; compliance; interpersonal and institutional trust; cooperation; shared sense-making; behavioral adaptation	Predominantly conceptual, qualitative, or embedded within broader organizational and socio-technical frameworks; limited standalone operational metrics	Literature- and expert-informed domains, organizational self-assessment, workforce and awareness criteria, and broader cognitive or social frameworks	Need to distinguish individual variables from collective, cultural, relational, and institutional constructs; limited direct operationalization and validation as cyber resilience measures	Linkov et al. [2]; Snyder et al. [12]; World Economic Forum [14]; Al-Hawamleh [29]

The indicator families shown in the table represent conceptual categories organized within the three a priori analytical domains rather than validated or standardized metrics. The domains are not mutually exclusive, and the representative sources are illustrative. The complete source-level classification is provided in Supplementary Table S1

propositions should not be interpreted as empirically validated causal relationships (Table 2).

4 Discussion: comparative positioning and conceptual contribution of ICRIM

The findings show that cyber resilience measurement is characterized not only by differences in indicator content, but also by differences in the level and form of operationalization. Technical approaches frequently provide explicit mathematical, temporal, structural, or probabilistic metrics. Organizational approaches more often rely on maturity criteria, structured assessments, expert judgment, or mission-oriented models. Socio-psychological and relational conditions are usually embedded within broader cognitive, social, workforce, or organizational structures rather than operationalized as independent cyber resilience measures.

This uneven development does not imply that one domain is inherently more important than another. Instead, it indicates that the three domains differ in the types of evidence through which they are operationalized and in the measurement procedures they require. The analytical problem is therefore not simply the absence of indicators, but the difficulty of relating heterogeneous indicator families without treating conceptual attributes, operational metrics, maturity assessments, and composite scores as equivalent.

4.1 Comparative positioning in relation to existing frameworks

Existing frameworks address important components of this problem, but they do so from different analytical and practical perspectives. The MITRE cyber resiliency publications provide a detailed systems-engineering structure connecting resilience objectives, techniques, design principles, metrics, measures of effectiveness, and context-specific scoring procedures [1, 4–7]. Their principal strength lies in operational depth at system and mission levels, although their primary orientation remains technical and engineering-based.

The Resilience Matrix developed by Linkov et al. [2] offers a broader multidimensional structure by combining four resilience stages—plan or prepare, absorb, recover, and adapt—with physical, information, cognitive, and social domains. It therefore constitutes an important socio-technical precedent for ICRIM. However, its measures are illustrative, and it does not systematically distinguish conceptual indicators from operational metrics or specify how constructs spanning several domains should be classified.

The World Economic Forum's Cyber Resilience Index provides an organization-oriented assessment structure based on measures of performance against resilience practices [14]. It gives greater attention to governance, leadership, workforce, ecosystems, and organizational improvement than predominantly technical frameworks. Its purpose, however, is organizational assessment and benchmarking rather than the conceptual classification of heterogeneous indicator and metric traditions.

The supplementary comparison also considered three institutional approaches that were not included in the 33-publication analytical corpus. NIST SP 800-160 Volume 2 Revision 1 provides systems security engineering guidance for achieving cyber-resiliency outcomes throughout the system life cycle [25]. The ENISA CSIRT Maturity Framework assesses the development of organizational, human, technological, and procedural capabilities within the specific context of national CSIRTs [26]. The CISA Cyber Resilience Review is an interview-based assessment of organizational operational resilience and cybersecurity practices [27]. These approaches provide established engineering, maturity-assessment, and operational-assessment procedures, but they serve purposes different from the conceptual synthesis undertaken in this study (Table 3).

4.2 Distinct conceptual contribution of ICRIM

The contribution of ICRIM does not lie solely in identifying technical, organizational, and socio-psychological and relational dimensions, since related distinctions already appear in socio-technical resilience research. Its contribution lies in combining four more specific analytical functions.

First, ICRIM maintains an explicit distinction between indicators and metrics. The model organizes the attributes and capacities that are relevant to resilience without assuming that all of them have already been translated into valid or comparable measurement procedures.

Second, it classifies heterogeneous contributions within a common analytical structure while retaining differences in their form of operationalization. Mathematical metrics, temporal measures, maturity criteria, expert assessments, behavioral constructs, and conceptual capabilities are therefore related without being treated as methodologically equivalent.

Third, ICRIM explicitly addresses constructs that cross domain boundaries. Training provision and formal communication structures, for example, are primarily organizational, whereas the resulting awareness, confidence, cooperation, or behavioral change may be socio-psychological or relational. Technical performance similarly acquires organizational meaning when interpreted in relation to mission requirements, continuity thresholds, and decision priorities. These cross-domain relationships are also shaped by the institutional and material conditions through which resilience is enacted, including technical standards, infrastructures, organizational routines, professional expertise, and the distribution of authority and decision-making power. Within

Table 3 Comparative positioning of ICRIM in relation to selected cyber resilience frameworks and assessment approaches

Framework	Primary orientation	Operational form	Treatment of domains	Relationship to ICRIM
MITRE Cyber Resiliency Framework and metrics publications	Systems engineering and mission resilience	Objectives, techniques, metrics, measures of effectiveness, and situated scoring	Primarily technical and mission-oriented, with contextual organizational relevance	Provides extensive operational depth; ICRIM offers a broader classification layer connecting technical indicators with organizational and socio-psychological or relational families
Linkov Resilience Matrix	Multidimensional socio-technical resilience	Four stages × four domains, with illustrative qualitative and quantitative measures	Physical, information, cognitive, and social	Provides a major socio-technical precedent; ICRIM distinguishes indicators from metrics and makes cross-domain classification and operationalization status more explicit
World Economic Forum Cyber Resilience Index	Organizational resilience and benchmarking	Measures of performance, scoring, and self-assessment	Governance, leadership, workforce, ecosystem, and technical practices	Provides an organizational assessment tool; ICRIM does not benchmark organizations but relates organizational measures to technical and relational indicator families
NIST SP 800-160 Vol. 2 Rev. 1	Systems security engineering	Cyber-resiliency objectives, techniques, approaches, and life-cycle guidance	System, mission, business, and stakeholder considerations	Provides implementation-oriented engineering guidance; ICRIM functions as an indicator-level conceptual structure rather than an engineering handbook
ENISA CSIRT Maturity Framework	CSIRT capability and maturity development	Self-assessment, peer review, and maturity levels	Organizational, human, technological, and procedural capabilities	Provides a context-specific maturity model; ICRIM has a broader conceptual scope and does not define maturity levels
CISA Cyber Resilience Review	Organizational operational resilience	Interview-based assessment of institutionalized practices	Governance, risk management, incident management, continuity, and related organizational practices	Provides an applied organizational assessment; ICRIM does not provide an interview instrument or maturity scoring procedure
ICRIM	Conceptual integration of cyber resilience indicators	Indicator taxonomy and cross-domain relational structure; no validated scoring procedure	Technical, organizational, and socio-psychological and relational	Organizes heterogeneous indicator families, distinguishes them from operational metrics, and identifies their cross-domain relationships and operationalization gaps

NIST SP 800-160 Volume 2 Revision 1, the ENISA CSIRT Maturity Framework, and the CISA Cyber Resilience Review were examined as supplementary comparative frameworks and were not included in the 33-publication analytical corpus

ICRIM, these conditions are not treated as a separate fourth domain; they operate as contextual and cross-domain mechanisms that influence how indicators are defined, interpreted, and acted upon. These examples reflect cross-domain patterns identified in the reviewed corpus. The specific relational structure through which ICRIM connects the three domains, however, is a theory-informed conceptual construction introduced by the authors rather than a relational structure directly reported or empirically validated across the reviewed studies.

Fourth, building on the cross-domain patterns identified in the synthesis, ICRIM proposes a feedback-oriented representation of the relationships among the three domains. Technical outcomes can inform organizational decisions; organizational processes can shape human understanding and action; and behavioral or relational responses can influence the implementation of procedures and technical controls. These relationships are not presented as empirically established causal effects. They provide a structure for future studies to define and test specific pathways, evidence requirements, and measurement procedures.

ICRIM should therefore be interpreted as a conceptual integration layer rather than as a replacement for MITRE, NIST, ENISA, CISA, WEF, or other established assessment approaches. Its purpose is to make explicit how indicator families drawn from different traditions can be located, distinguished, and related before context-specific metrics, weights, thresholds, or scoring systems are developed.

4.3 Research, practice, and regulatory implications

For research, ICRIM provides a basis for developing domain-specific and cross-domain operational measures. Future studies could specify observable variables for each indicator family, identify suitable evidence sources, examine construct validity, and test whether proposed relationships remain stable across sectors and organizational settings. Particular attention is required for socio-psychological and relational constructs, where the corpus provides less evidence of standalone operationalization.

For practice, the model can be used as a conceptual checklist for identifying whether an assessment concentrates disproportionately on system performance, organizational capability, or human and relational conditions. At its current stage, however, it should not be used to calculate an overall resilience score, rank organizations, or make claims of comparative performance.

ICRIM may also support future mapping to regulatory and standardization frameworks. For example, governance, risk-management, continuity, incident-management, training, communication, and learning requirements could be associated with relevant technical, organizational, and relational indicator families. Such a mapping could help distinguish formal compliance requirements from the evidence needed to assess the underlying resilience capabilities. Nevertheless, the present study does not conduct a clause-by-clause mapping to the NIS2 Directive or ISO/IEC 27035-1:2023 and therefore makes no claim of formal alignment, compliance, or improvement over these frameworks [30, 31].

Overall, ICRIM addresses conceptual fragmentation by providing a traceable structure for organizing different types of cyber resilience indicators and examining their relationships. Its value at this stage is analytical rather than predictive or evaluative. Empirical validation, operational metric development, weighting, scoring, and comparative testing remain necessary before the model can support standardized organizational assessment.

5 Conceptual schema of the integrated cyber resilience indicators model (ICRIM)

Figure 1 presents the Integrated Cyber Resilience Indicators Model (ICRIM) as a non-hierarchical and dynamically interconnected structure comprising three domains: technical, organizational, and socio-psychological and relational. The three domains are positioned at an equal distance around the central concept of cyber resilience, indicating that none constitutes the primary or foundational layer of the model. Instead, ICRIM represents the three domains as interdependent components that should be considered jointly when organizing and interpreting cyber resilience indicators.

The technical domain includes indicators associated with system performance, detection, response, retained functionality, availability, redundancy, recovery, and operational continuity. These indicators describe how systems and infrastructures behave under adverse cyber conditions. Their significance, however, depends on the organizational priorities, mission requirements, and decision thresholds through which technical performance is interpreted.

The organizational domain includes governance responsibilities, resource allocation, preparedness, incident-management processes, communication structures, business continuity, training provision, workforce capability, and organizational learning. It represents the institutional mechanisms through which technical information is translated into coordinated decisions and resilience practices.

The socio-psychological and relational domain includes awareness, risk perception, confidence, protective behavior, trust, cooperation, shared sense-making, and behavioral adaptation. These indicators concern how individuals and groups understand, accept, and implement technical and organizational resilience measures. Formal governance arrangements, organizational culture, and communication structures remain primarily organizational constructs,

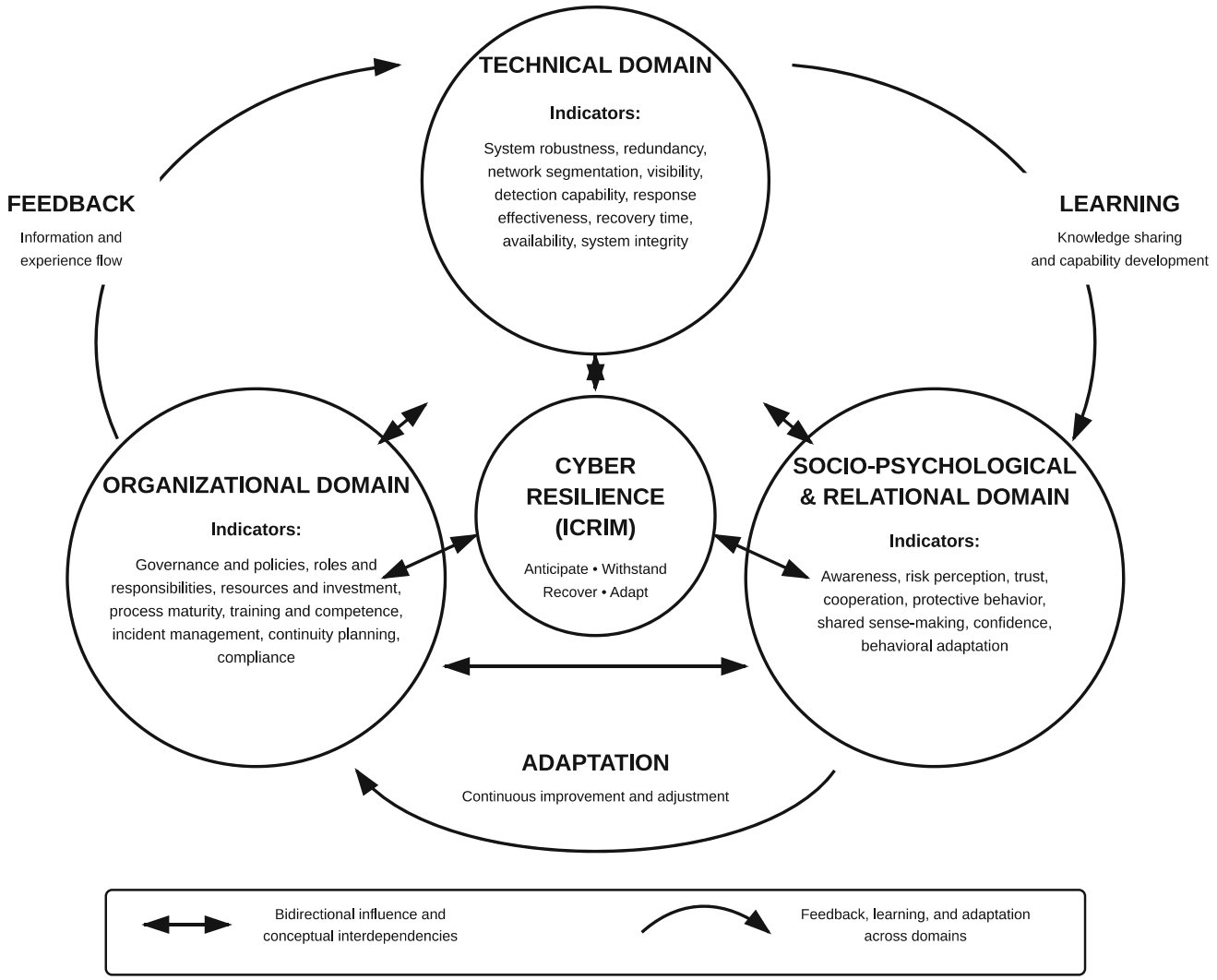


Fig. 1 Integrated Cyber Resilience Indicators Model (ICRIM). The model illustrates three interdependent indicator domains and the conceptual feedback, learning, and adaptation processes through which their relationships are represented

although their effects may be reflected in behavioral and relational responses.

The bidirectional links among the three domains represent their conceptual interdependencies. Technical outcomes can influence organizational decisions and resource allocation. Organizational practices can shape awareness, trust, cooperation, and protective behavior. Human and relational responses can, in turn, affect the implementation of procedures and the effective use of technical controls. The circular flow surrounding the model represents feedback, learning, and adaptation as conceptual processes linking the three domains over time.

The bidirectional relationships shown in Fig. 1 should not be interpreted as empirically validated causal pathways or as relationships directly reported across the reviewed sources. Rather, they are theory-informed conceptual propositions introduced within ICRIM to organize the cross-domain interdependencies identified through the synthesis. They provide a basis for future operationalization and empirical testing. Their specific expression may vary according to the sector, organizational setting, mission requirements, threat environment, regulatory context, and available evidence.

ICRIM therefore does not prescribe universal metrics, weights, thresholds, or scoring procedures. Its purpose is to organize heterogeneous indicator families within a coherent socio-technical structure, clarify their areas of overlap, and support the future development of context-appropriate measurement approaches. The model should consequently be understood as a conceptual integration framework rather than as a validated assessment instrument.

6 Conclusions and future research

This study examined how cyber resilience has been conceptualized and operationalized across an analytical corpus of 33 publications issued between 2011 and 2024, identified through the search and relevance-screening process. The analysis organized the reviewed indicator families within three interdependent domains: technical, organizational, and socio-psychological and relational. It also showed that these domains differ substantially in their degree and form of operationalization. Technical approaches in the corpus typically employ explicit mathematical, temporal, structural, or probabilistic metrics. The complete source-level classification of the analytical corpus, including sources [32–43], is provided in Supplementary Table S1.

The study proposes the Integrated Cyber Resilience Indicators Model (ICRIM) as a conceptual structure for organizing these heterogeneous indicator families. Its contribution lies in distinguishing indicators from metrics, preserving differences among measurement traditions, and making cross-domain relationships more explicit. ICRIM does not provide a validated measurement instrument, scoring procedure, weighting system, or universal set of operational metrics. It should therefore be interpreted as an analytical framework intended to support subsequent operationalization rather than as a completed assessment model.

Several limitations should be considered. First, the analytical corpus resulted from the search and relevance-screening process and does not constitute an exhaustive or bibliometric representation of all cyber resilience research published during the period examined. Second, source extraction and primary classification were conducted by the first author and were not subjected to independent double coding or statistical assessment of inter-coder agreement. Third, the three-domain structure was used as an initial theory-informed analytical lens, and alternative classifications of cross-domain constructs may remain possible. Finally, ICRIM has not been empirically validated through real-world application, simulation, comparative organizational assessment, or longitudinal testing.

Future research should therefore focus on the operational development and evaluation of the proposed structure. This includes defining observable variables and evidence requirements for each indicator family, developing context-appropriate metrics, and examining the reliability and validity of proposed measures. Particular attention is required for socio-psychological and relational indicators, where the reviewed literature provides less evidence of standardized operationalization.

Further work should also investigate how indicators from different domains can be combined without obscuring their distinct methodological properties. This requires testing alternative aggregation, weighting, and scoring procedures and determining whether cross-domain relationships remain stable across sectors, organizational settings, mission requirements, and threat environments. Empirical studies could also examine how technical outcomes, organizational capabilities, and human or relational responses interact during preparation, disruption, recovery, and adaptation.

Finally, future research may explore how ICRIM can be mapped to sector-specific assessment frameworks, regulatory requirements, and organizational resilience practices. Such work should distinguish clearly between conceptual correspondence, operational compatibility, and formal compliance. At its current stage, ICRIM provides a structured basis for these further developments, while its practical applicability and measurement validity remain to be established through empirical research.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10207-026-01341-y.

Author contributions All authors contributed to the writing of the manuscript and reviewed the final version.

Funding Open access funding provided by HEAL-Link Greece.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical standards This article does not contain any studies with human participants or animals performed by any of the authors.

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